Water Conservation Science and Engineering (2026) 11:1
https://doi.org/10.1007/s41101-025-00467-9

REVIEW

Advances in Water Footprint Research: Trends, Methods, and Implications - A Literature Review

Zhenna Yi[1] · Chengyan Li[1] · Yahong Dong[1,2]

Received: 1 July 2025 / Revised: 1 December 2025 / Accepted: 5 December 2025
© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2025

Abstract

The water footprint (WF) has become a vital tool for water resource management, attracting growing scholarly interest. This study reviews WF articles from 2000 to 2023, exploring development trends, geographical distribution, research focuses, methodologies, implications, and policies. It is found that three key cross-cutting barriers persist in WF research: issues with data consistency and transparency, basin- and time-sensitive impact attribution, and limited policy uptake. In practice, WF analysis informs urban demand management, industrial permitting, and product policy through life-cycle reporting and disclosure. Geographical distribution indicates that China, with 588 papers, and the United States, with 221, are the two countries with the highest number of publications. The significant number of publications from China can be attributed to the recent emphasis on agricultural water conservation. Two primary approaches to WF assessment are compared: the volume-oriented WF approach (VWFA) and the impact-oriented WF assessment (IWFA). Advancements can be made through scarcity-weighted indicators, improved Multi-Regional Input-Output Model (MRIO)-based tracing, and basin-level interventions. Considering global food security, this review highlights how countries with large WFs can export virtual water. It clarifies capabilities and limitations of WF methods, offering steps to align scientific accounting with policy and management.

Keywords Water footprint · Life cycle assessment (LCA) · Bibliometric analysis · Sustainable development · Water scarcity

Introduction

With the rapid growing population and the significant impacts of climate change, water scarcity has become an increasing challenge [1–4]. The water footprint (WF) provides a measure of water use by quantifying the total volume of freshwater used for various human activities [5], including direct withdrawals such as domestic use by households and municipal services, and indirect use through water embodied in products and services [6]. WF comprises three components: green (rainwater), blue (surface and groundwater), and grey (pollution assimilation) [7]. The green WF accounts for crop evapotranspiration within the product system boundary. Surface and ground water in natural ecosystems not used for production is part of the basin water balance but is excluded from a product's green WF. The grey WF represents the theoretical volume of freshwater required to dilute a pollutant load from an activity to meet a specified ambient quality standard. It does not represent a physical withdrawal or hydrological "loss". In

Zhenna Yi and Chengyan Li contributed equally to this paper and should therefore be acknowledge as co-first authors.

✉ Yahong Dong
yhdong@must.edu.mo

Zhenna Yi
3240009055@student.must.edu.mo

Chengyan Li
richardlcy@foxmail.com

[1] Department of Environmental Science and Engineering, Faculty of Innovation Engineering, Macau University of Science and Technology, Macao 999078, China

[2] National Observation and Research Station of Coastal Ecological Environments in Macao, Macau University of Science and Technology, Macao 999078, China

Published online: 20 December 2025

reporting, consumptive use (evaporation and incorporation) is distinguished from return flows, with grey WF representing water-quality pressure rather than volumetric loss.

WF can support water resource management by highlighting areas of high stress and by structuring information for allocation decisions when aligned with basin governance and data quality [8]. It also provides a compact typology of water risks faced by industry. These risks fall into four categories: physical scarcity, water quality constraints, regulatory constraints and reputational concerns. Addressing these risks enables industries to reduce blue water use through lower consumption and to reduce grey water pressure through improved effluent control, which contributes to more stable production during dry seasons. Corporate ESG (environmental, social, and governance) reporting frameworks are driving more consistent disclosures of water use, basin context, and supply-chain risks [9]. As water shortages intensify, WF provides a valuable framework for structuring evidence, given that results are interpreted with consideration of basin context, trade structures, and social needs [10].

The theory of WF is rooted in the concept of virtual water, introduced by Allan in 1993 [11], and later refined by Hoekstra [5], who formalized the water footprint concept. Chapagain and Hoekstra [12] developed the first product-level metrics, and the Water Footprint Network (WFN) subsequently published the Water Footprint Assessment Manual [7]. ISO 14046 [13] introduced a Life Cycle Assessment (LCA)-based method. Two distinct methodological approaches are often adopted. The volume-oriented WF approach (VWFA) quantifies green, blue, and grey water footprints as volumes (m^3) along supply chains, with grey water serving as a theoretical proxy for the volume needed to dilute pollutants to meet a specified ambient quality standard. In contrast, the impact-oriented WF assessment (IWFA) evaluates potential impacts, such as water scarcity, using regionalised characterisation factors.

In recent years, there has been a significant increase in WF research. The research topics include calculation methods, applications in different sectors, and implications for policy and practice. A bibliometric analysis on 639 papers on the WF published from 2006 to 2015 indicated a substantial growth trend over the past years [14]. Key contributions include the foundational work of Globalization of Water, which introduced the concept of virtual water trade and its implications for global water security. Mekonnen and Hoekstra [8] provided a comprehensive assessment of the WF of humanity, quantifying global water use patterns and highlighting hotspots of water consumption and pollution. The Water Footprint Assessment Manual established the global standard for conducting WF studies, offering a methodological framework that has been widely adopted [7]. National and sectoral studies, such as analysis of national WF [15] and calculation of WF of farm animals and crops [16], have further detailed its applications. Recent research has increasingly concentrated on specific regions and topics. For example, Zhuo et al. [17] studied China's WF and virtual water trading. Hogeboom et al. [18] integrated WF with other environmental indicators. Zhu et al. [19] mapped research landscapes based on a bibliometric analysis, identifying trends and gaps in the literature. Additionally, Liu et al. [20] explored water use efficiency in crop production, while Pfister et al. [21] and Gerbens-Leenes et al. [22] applied WF to studying specific products such as cotton and construction materials. WF can also be linked to broader sustainability issues, as discussed in Ercin and Hoekstra [23] and Hoekstra et al. [7]. WF can be applied to other study cases, including Mekonnen and Hoekstra [24] on animal products, Hoekstra [25] on consumer society, Vanham et al. [26] on Italy's water scarcity, and Chapagain et al. [27] on cotton consumption.

The rapid growth of WF research highlights the need for a systematic synthesis. Although reviews like Pfister and Bayer [10] provided valuable insights into the strengths and limitations of WF, their analysis summarised recent advancements. Specifically, their work noted the conventional WF's limitation in capturing the spatial and temporal variability of water impacts, thereby preventing its policy utility. Bibliometric analyses by Zhang et al. [14] and Zhu et al. [19] have mapped the research development of WF but have not investigate deeply into the methodological nuances or the practical applications of WF assessments. Other reviews, such as Hoekstra [28], provided broad insights but typically focused on specific topics like sustainability or global challenges. In addition, previous reviews noticed the limitations of WF, i.e., the lack of standardised methodologies, challenges in integrating green water use, and insufficient consideration of equitable water sharing [28, 29]. However, there is a lack of review articles that synthesise the two WF approaches, namely VWFA and IWFA, nor is there a critical evaluation of the different WF assessment methods. Additionally, the potential of these insights to inform more effective management strategies in the face of escalating water scarcity and environmental challenges remains underexplored [20, 30–32].

This systematic review aims to address existing gaps in WF research by synthesising the latest literature, critically evaluating various approaches to WF assessment, and exploring how these insights can inform more effective management strategies in the context of growing water scarcity and environmental challenges. The review focuses on two established accounting frameworks: VWFA and IWFA. It examines recent trends, methods, and implications in WF research. Furthermore, the study reviews the applications

and policies relevant to WF research, discusses future practices, and elucidates the significance of WF studies for water management.

Foundation and Evolution of the Water Footprint

The Concept of the Water Footprint

The concept of WF was proposed by Hoekstraand and Hung [5], which can effectively provide a scientific basis for water resource management. Their study showed that international trade transfers "virtual water," so water-scarce regions can relieve local pressure by importing water-intensive products. They demonstrated this by estimating crop-specific virtual water contents from crop water requirements (green and blue) and combining these with bilateral trade statistics to derive national virtual-water flows. This work used an accounting approach rather than the LCA approach as given in ISO 14046. WF quantifies water use through a supply-chain perspective. It links direct and indirect water consumption, enhances the traditional method of calculating water consumption, and reflects the actual water use of human activities [28, 33]. Indirect water consumption refers to the virtual water embodied in goods and services along the supply chain. According to the Water Footprint Assessment Manual [7], "water footprint is an indicator of freshwater use that looks not only at direct water use of a consumer or producer, but also at the indirect water use". Thus, WF differs from conventional abstraction metrics by encompassing blue, green, and grey components, offering a composite measure of water appropriation.

The concept of WF has been applied in multiple contexts, though the extent and consistency of application vary across regions and sectors. The 2020 United Nations Water Development Report [4] uses WF as an indicator for energy production, in water use per unit of energy (e.g., m^3/kWh), distinguishing results by technology and context to enable consistent comparisons across pathways. WF measures water consumption and relates it to economic efficiency by tracking flows within study objects, such as basins, products, sectors or companies, which are chosen to keep system boundaries coherent and to match the decision scale [34, 35]. On the other hand, context-specific options are evaluated using common metrics, including water intensity (m^3 per unit), scarcity-weighted results and compliance with water-quality standards, so results can remain comparable. Unlike traditional water abstraction indicators, WF is flexible and can be quantified at multiple scales, from products and plants to business and individuals, and up to regions and nations, using consistent definitions and boundary rules [25]. When utilised in conjunction with basin governance and real-time allocation, WF offers a structured framework that links water consumption to economic activity, facilitating the comparison of various options across different locations and time periods.

Table 1 Suggestions of WF reporting for WF sources and study objects [7, 28]

Category	Item	Suggestions for reporting
WF source	Green water	Water stored in soil from rainfall and used by plants via evapotranspiration.
	Blue water	Surface and groundwater consumed.
	Grey water	Theoretical assimilative volume to meet the ambient standard; state pollutant, standard and background.
	Product water footprint	Blue, green and grey separately; state boundary.
	Personal or group water footprint	Blue, green and grey separately; state boundary; adding per-capita if relevant.
	Business water footprint	Blue, green and grey; splitting operational and supply chain.
Study object	Regional/national water footprint of production	Blue, green and grey generated within the territory in the accounting year; virtual water export reported separately; imports excluded.
	Regional/national water footprint of consumption	Internal and external WFs for goods and services consumed by residents; internal inside the territory; external outside the territory for imports; exports excluded; report net external balance when possible.

Entries are volume-oriented under VWFA and use cubic metres. Impact-oriented metrics under ISO 14046 should be labelled and reported separately in cubic metre equivalents

Components of the Water Footprint

WF encompasses virtual, domestic, and ecological waters, including that contained in the products and services utilised by humans [15]. Table 1 summarises the main WF variants. Hoekstra [28] categorises into three contributing factors: the green WF, the blue WF, and the grey WF for a defined product, sector, or region. Green water counts the share of rainfall evapotranspiration by the managed crop or vegetation over the production period (for example, the growing season or the reporting year). Rainfall that runs off or percolates is not green water, and non-productive evapotranspiration from weeds is excluded unless it is part of the managed system, while green and blue water components are reported separately to avoid double counting. The blue WF indicates the volume of surface and groundwater used along a product's supply chain. The grey water is the equivalent volume needed to dilute a pollutant load to the regulatory standard. To calculate grey WF, pollutants like COD

and TN loads from monitoring or inventories can be used, with the legal thresholds for those pollutants, and ambient background concentrations [36]. Grey WF is reported separately from blue and green consumptive use, so it is not registered as consumption of whichever water body provides dilution [8].

In addition, WF can be categorised according to the study object, including product WF, personal or group of consumers WF, WF of a business, regional water footprint, and national WF [7]. Hoekstra et al. [7] also described each of these categories. A product WF is the sum of blue, green, and grey water used along its production chain, including indirect water in the supply chain [37]. It is the basic unit of water embodied in products. In business accounts we report blue water and green water separately and as a total [38]. On-site use is partitioned by source, so the blue and green components remain clear and consistent with the set nomenclature. The personal or group of consumers WF reflects the amount of blue, green, and grey water required associated with the goods and services consumed by individuals or groups of consumers, as well as the water footprint contained within the products consumed [39]. An individual WF is the total amount of blue, green, and grey water required by a person who uses goods and services. WF of a group of consumers equals the sum of WFs of individual consumers. The business WF is defined as the total amount of blue water and green water used directly or indirectly to operate and support the business. Its supply-chain (indirect) WF is the blue, green, and grey water required to produce all input goods and services [38].

The regional WF is the total water consumption and pollution that occurs within a geographic area, which may be a watershed, province, state, or other spatial unit [40]. Regional WF studies provide a basis for developing water resource strategies that address specific local water resources and consumption characteristics. A national WF includes water use and pollution within a country plus virtual water embodied in imports and exports [41]. The composition of the national water footprint consists of an internal WF and an external WF. The internal WF is the part of a country's consumption footprint that occurs inside its territory [40]. It is the water used domestically to produce goods and services that residents consume. A larger internal share signals stronger water self-reliance. The external WF is the part of the consumption footprint that occurs outside the territory to produce imported goods and services that residents consume [41]. By contrast, the production WF shall cover all water used within the territory to produce goods and services in the accounting year, and it includes the water embodied in exports, which should be reported as virtual water export. A larger external share indicates greater dependence on trading partners and is useful for assessing water security and equity.

Historical Development

The WFN approach reports volumetric results (m^3) for allocation and conservation, while ISO 14046 reports impact-oriented results (for example, scarcity-weighted units) for product system comparison [42, 43]. Policymakers should align the framework with the specific decision at hand: using the WFN approach for basin allocation and conservation planning, and ISO 14046 for product comparison and disclosure, and when feasible, both should be reported with clear labelling [44, 45]. However, the coexistence of both approaches may cause confusion. To mitigate this risk, a transparent framework declaration and side-by-side reporting are recommended.

Given that agriculture accounts for approximately 70% of global freshwater withdrawal [46], it has long been recognised as a priority area for investigation within the WF analysis. Hoekstra and Hung [47] were among the first to produce global estimates of water consumption for various crops across multiple countries, though their study did not distinguish among green, blue, and grey water components. Numerous studies, for instance, have examined WFs of major crops such as wheat, maize, and rice [48, 49]. These studies highlight the significant variation in WFs across different regions, influenced by factors such as climate, soil type, and agricultural practices [22]. Such analyses reveal hidden water uses that are not visible to end users in food production, highlighting the need for more sustainable water management in agriculture. Current research in agricultural WFs centres on three main themes. The first area explores the spatial and temporal evolution of agricultural WFs, comparing these footprints across regions to generate new insights for agricultural water management [34, 50, 51]. Beyond on-farm production, water is also used in post-harvest stages such as processing, storage, distribution, and retail. In cases where detailed inventories are unavailable, these stages are acknowledged qualitatively in the system boundary notes. The second area of research focuses on assessing water stress and water use efficiency based on crop WFs, with the goal of optimizing agricultural planting structures [52–54]. The third area investigates the factors that influence the crop WFs [55, 56].

As the concept of WF has gained wider recognition, its applications have expanded from agriculture to encompass industrial and domestic sectors. Industrial WF has become a significant area of research due to the considerable volume of water utilised in manufacturing processes [57–60]. For example, WF of the textile and apparel industry, which is known for its high water consumption and pollution, has

been subjected to close scrutiny [61]. These studies emphasise the necessity for industry-specific strategies to reduce water consumption and mitigate environmental impacts. Additionally, energy production and basic manufacturing industries have been the subject of WF research. For instance, the spatial characteristics of energy-water use efficiency and WFs can be quantified, thereby facilitating a more comprehensive understanding and providing valuable insights for energy and water resource management decisions [62]. Furthermore, the grey WFs generated in the production of energy have been examined, and the potential impacts on aquatic ecosystems have been investigated [63]. For thermal power plants, the cooling technology (once-through vs. recirculating) and report withdrawals and consumption can be specified separately. Other energy systems (e.g., hydropower, bioenergy, PV, and wind) have distinct water profiles and shall be reported accordingly. While WF accounting is not a direct measure of service access, it complements water supply and sanitation planning by quantifying water demands and pollutant loads at the city scale. Similarly, studies of WFs encompass both household water use (e.g., drinking, cooking, and cleaning) and non-domestic public water use in urban areas, along with an ecological recharge component [64]. A regional WF study typically includes assessments of general household water use, which are then aggregated to calculate the national or regional WF [65, 66].

Introducing the concept of WF was a significant milestone in water resources management. From its inception as a means of quantifying water use in agriculture to its present position within the global discourse on sustainable development, WF has furnished invaluable insights into the intricate dynamics of water use and its consequences for the environment and society. WF analysis supports Integrated Water Resources Management (IWRM) by linking demands, basin context, and pollution thresholds [28]. These efforts can also contribute to SDG 6 (clean water and sanitation) by connecting efficiency and water-quality objectives with actionable solutions [67]. Expanding data sharing and incorporating case studies from under-represented regions are critical to the wide application of WF. Therefore, this study integrates a systematic review with targeted critical appraisals of calculation methods, practical implications, and relevant policies and standards.

Research Method

This study combines both systematic review and critical review. This first part is a bibliometric analysis based on a systematic review. The articles were collected through the Web of Science (WOS) database. We conducted the search in Oct 2024 by inputting the selection string of AK= ("water footprint"), and the scope of screening was "topic" to search in title, abstract and keywords. The time frame was from Jan 01, 2000, to Dec 31, 2023, to cover over two decades of possible publications. The document types considered in this study were "article".

The screening proceeded in two stages. In the first stage, titles and abstracts were reviewed against predefined rules. Inclusion criteria required peer-reviewed journal articles that quantify or characterise WF or virtual water at the product, process, facility, sector, regional, or national scale. These articles must define system boundaries, specify the green, blue, and grey components or ISO 14046 impact categories, and document data sources. Exclusion criteria removed studies that merely mention the concept without providing calculations or characterisations, those outside the scope (e.g., hydrology or life cycle assessments without WF indicators), non-peer-reviewed sources, and duplicates. In the second stage, full texts were reviewed when abstracts were insufficient to determine eligibility. The records retained after screening constituted the included set for analysis. A bibliometric analysis was conducted to examine development trends, geographical distribution, and research focus areas. For the critical review, an in-depth evaluation of calculation methods, practical implications, and relevant policies and standards was performed. A total of 1,625 articles were obtained (see the Supplementary Material).

Results and Discussion

Research Trend

As illustrated in Fig. 1, there is a significant increase in the number of publications pertaining to WF from 2000 to 2023. The pattern of the number of publications per year, as summarised by WOS, allows for the description of WF research as comprising three phases [68, 69]. The rise in publications likely reflects both growing attention to the topic and a larger research community.

In the initial phase (2000–2008), the database contained few articles, and the concept of WF was still in its nascent stages. During this period, scholars began to examine global freshwater allocation and management issues and used WF concept to assess water resource sustainability [70].

In the second phase (2009–2017), the number of publications rose sharply, and the focus shifted from theory to practical applications. As societal awareness of the need to protect water resources increased, there was a corresponding increase in the number of articles concerned with the sustainable development of these resources [71]. Moreover,

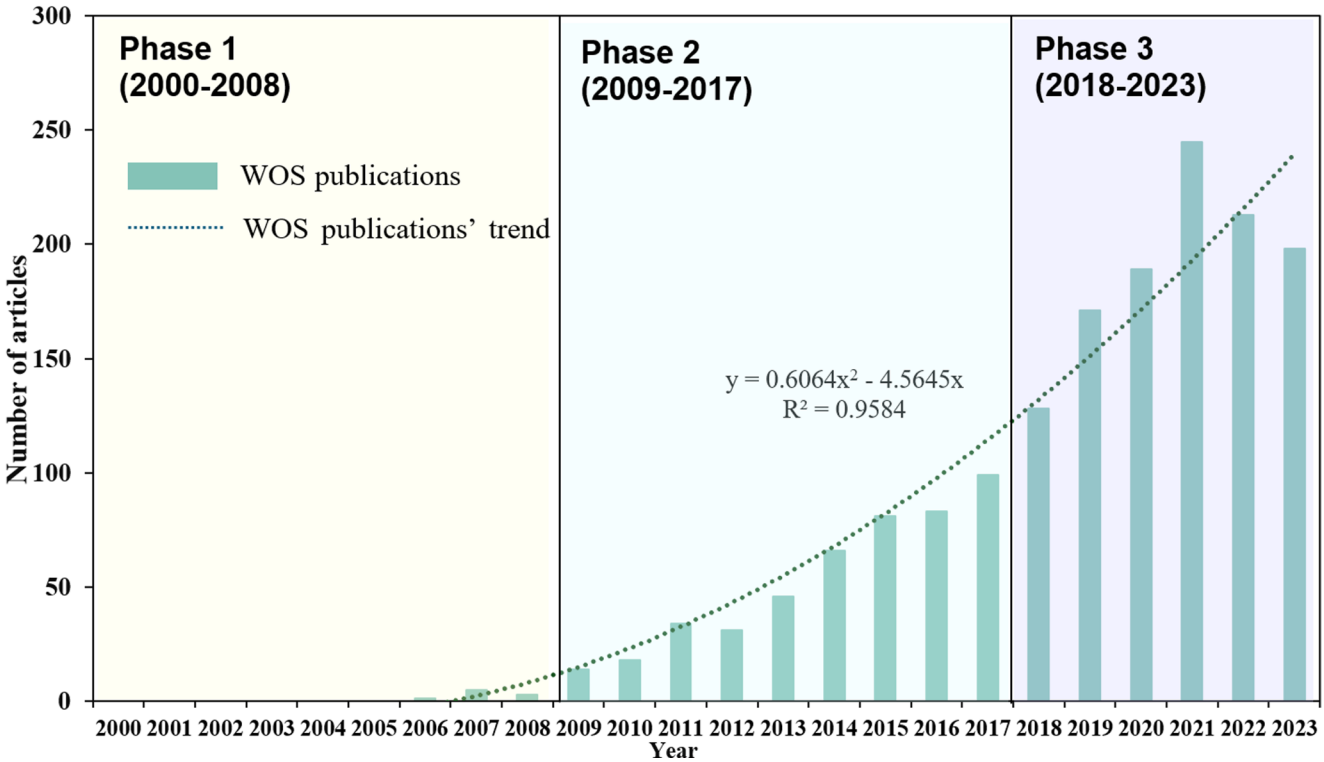
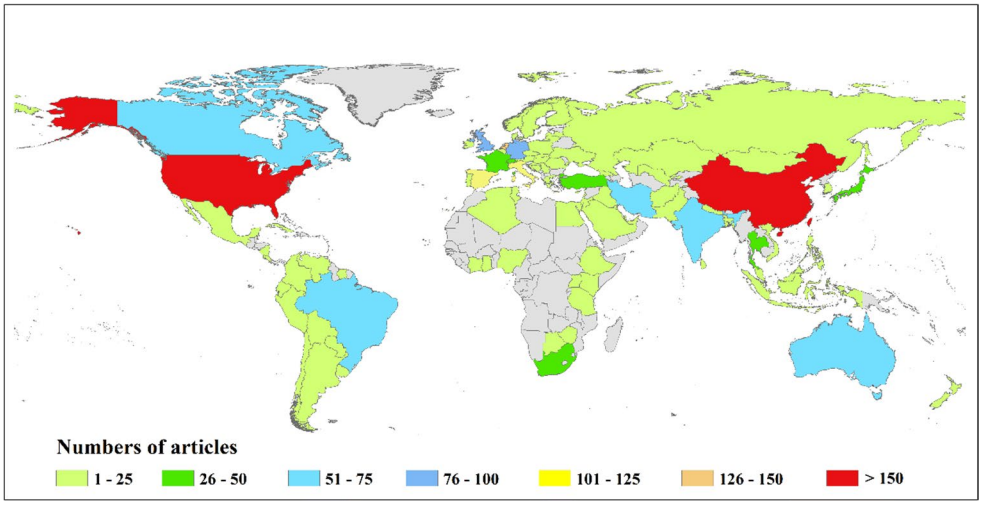


Fig. 1 The number of articles published from 2000 to 2023 and three development phases of WF

Fig. 2 Global distribution of WF publications by country



the standardised contribution of WFN and ISO to WF evaluation during this period has, to some extent, facilitated the global recognition of WF [72].

In the third phase (2018–2023), the literature demonstrates a fluctuating pattern of growth. In this phase, WF has been gradually becoming an effective tool for addressing water security and water allocation issues [73]. These studies position WF to inform city-scale supply-demand balancing, municipal efficiency measures, and risk and resilience questions, indicating urban water security as a growing emphasis. Recent studies on urban water management and city-scale water footprints indicate growing attention to urban water security [74, 75].

Spatial Distribution of Water Footprint Research

Regarding country literature statistics in WOS in Fig. 2, China has the highest number of contributions to the literature of WF, with 588 articles, followed by 221 articles from the United States. Additionally, the affiliations of lead authors and corresponding authors were coded to determine whether China-focused studies were led

by China-based researchers; the counts are provided in the Supplementary Material. As a prominent agricultural nation, China plays a key role in WF research, particularly in the agricultural sector [76]. Consequently, considerable attention has been directed toward understanding China's agricultural WF. The country's uneven distribution of water resources has led to water scarcity in several regions [77]. Using China as an example, publication volume is high, but water management has been uneven across regions and sectors, the reasons are not primarily scientific but institutional and economic [78]. Key gaps include harmonising data across provinces and sectors, incorporating grey-water pressure limits into discharge permitting, and creating incentives for efficiency and reuse [79, 80]. To address water scarcity, policy and investment actions are required, including the integration of footprint inventories into river-basin allocation and drought management plans, the alignment of tariff and permit reforms with benchmarks for blue, green, and grey water, the prioritisation of funding for reuse and conveyance where marginal abatement costs are lowest, and the reporting of dual metrics within a hydroclimatic and trade context [81, 82]. Similar strategies are applicable to other regions with varying governance structures [83]. Consequently, there has been a growing focus among scholars on studies of China's water footprint. As WF research becomes increasingly advanced, its emphasis is shifting towards the complex interrelationships between resources, society, and the economy [19]. Despite advances in WF research, the adoption of a unified accounting standard has been slow. Three key factors lead to this delay. First, data and boundary heterogeneity across provinces and sectors complicates harmonisation efforts [84]. Second, estimates of grey water depend on ambient standards and background concentrations, which vary by jurisdiction, thus hindering comparability [85]. Third, institutional responsibilities for allocation, quality control, and reporting are divided across multiple agencies, preventing the establishment of a single, standardised protocol [86]. A practical solution is a dual-track approach: at the international level, it is crucial to contribute to the development of shared characterisation factors and open datasets, ensuring that results are comparable across borders. Building such global capacity reduces the risk of divergent national rules being perceived as conflicts and aligns with the interests of both importing and exporting regions.

Agricultural and Industrial Water Footprints

After screening, a total of 499 agricultural WF articles and 254 industrial WF articles were retained. The results are presented in Fig. 3. A review of the literature on WF reveals that the first widely cited agriculture-focused journal article appeared in 2006 [87], followed by several studies in 2007 on other agricultural commodities such as coffee and tea [27]. In the included set for the period 2000 to 2023, studies on agricultural WF outnumber those on industrial WF. These counts reflect the studies retained after screening,

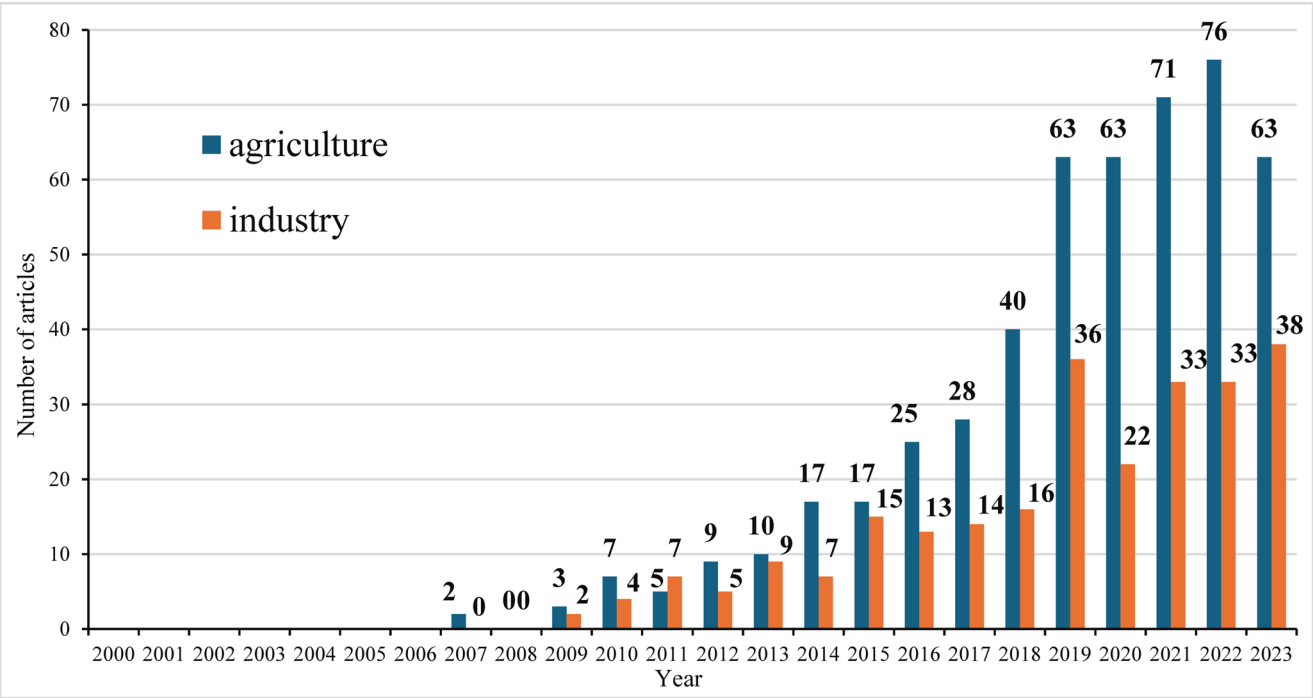


Fig. 3 Number of publications on the topic of agricultural and industrial WFs (2000–2023)

with 2000 to 2023 being the search period, and the earliest agricultural paper in the corpus dating to 2006. Over this period, publications on agricultural WF grew rapidly, while those on industrial WF increased at a slower pace.

It is important to note that the differentiation of WF studies based solely on industry classification remains somewhat limited in scope. The preceding searches were primarily focused on investigating the alterations and prospective capabilities of WFs in the advancement of the industry from a macro perspective. The current research on WFs is focused on two key areas: virtual water trade and WF of products [88, 89]. This is occurring concurrently with a notable shift in the global perception of freshwater as a scarce resource, driven by the growth of international trade in water-intensive goods. A study by Sun et al. [90] indicates that the most prevalent keywords in the field of WF research in the existing literature as of 2021 were water scarcity, carbon footprint, water depletion, and grey water. Beyond quantifying WFs, recent studies link WF with carbon accounting, climate risk and sustainable development planning [18]. Some studies paired blue, green and grey water indicators with greenhouse-gas inventories, energy use and exposure to climate variability to examine cross-resource trade-offs [91]. Its practical value is that it enables joint indicators and prioritisation rules to identify synergies and conflicts across water, carbon and resilience objectives, rather than treating each metric in isolation.

Domestic Water Footprints

Notwithstanding the growing number of water footprint studies, there remains a lack of regional or city-specific studies [92]. The study of urban WF is a crucial element in the management of water resources, which is essential for the construction of sustainable drainage systems in urban areas [93]. For example, the Guangdong-Hong Kong-Macao Greater Bay Area (GBA) is characterised by high population density, with Macao having one of the highest population densities in the world. A large population inevitably results in a considerable urban WF [94, 95]. Domestic WFs are primarily applied in local case studies to estimate household and service-sector water use within the defined study boundary [7]. At the city scale, studies calculated WF to assess residents' dependence on available supplies under the city's boundary and supply portfolio, while in many urban cases, the domestic component forms a substantial share of the consumption footprint [26]. Across our 2000–2023 search window, studies that analyse domestic WFs are most often embedded in multi-sector assessments alongside agricultural, industrial or environmental components, to compile regional or city-scale accounts and to inform urban water management [96].

Assessment Methods for Water Footprint

In recent years, the assessment and accounting methods of WF have become a significant area of focus in the context of the growing global attention to water resources management. Currently, WF assessment methodologies can be classified into two types: the volume-oriented water footprint assessment (VWFA) proposed by the WFN, and the impact-oriented water footprint assessment (IWFA) first proposed by Pfister et al. [21] and later summarised in the ISO 14046 standard. The specific differences between the two types are illustrated in Fig. 4.

The VWFA is primarily based on the WFN publication, The Water Footprint Manual, which outlines the methodology for setting target ranges, WF accounting, WF sustainability assessment, and formulating WF response plans [40]. A WF response plan translates accounting and assessment results into time bound actions for allocation, efficiency and pollution control. To prepare such a plan, decision makers require a baseline WF with uncertainty, information on water availability and quality constraints, projected demand and supply under climate scenarios, cost and feasibility of measures such as conservation, reuse and process change, and the trade and virtual water implications of sourcing shifts [97].

VWFA benchmarks green, blue and grey volumes against local water availability and quality standards [81]. VWFA provides an accounting of green, blue and grey volumes and can inform interpretation of the status of water resources when paired with hydrological context, including long term availability, seasonal conditions and recent anomalies. In practice, this assessment should be supplemented with indicators of dryness or wetness at the basin scale, such as precipitation and runoff relative to long term averages, storage levels and allocation status. It categorises water usage into three distinct groups for accounting purposes, facilitating an understanding of the impact of human activities and products on water scarcity and pollution. Furthermore, it offers solutions to ensure the sustainable utilisation of freshwater.

The IWFA is primarily concerned with conducting WF assessments in accordance with the standards set forth in ISO 14046 [44, 45]. This encompasses four fundamental components: the initial determination of goal and scope of the assessment, the subsequent analysis of the WF inventory, the assessment of sustainability or potential impacts of water use, and interpretation of results [13]. IWFA characterises potential impacts such as the water scarcity footprint and related indicators for human health and ecosystem quality.

The IWFA method may obfuscate the boundaries between the three categories of WFs. Similarly, as with LCA, the objective is to quantify the impact of water resource

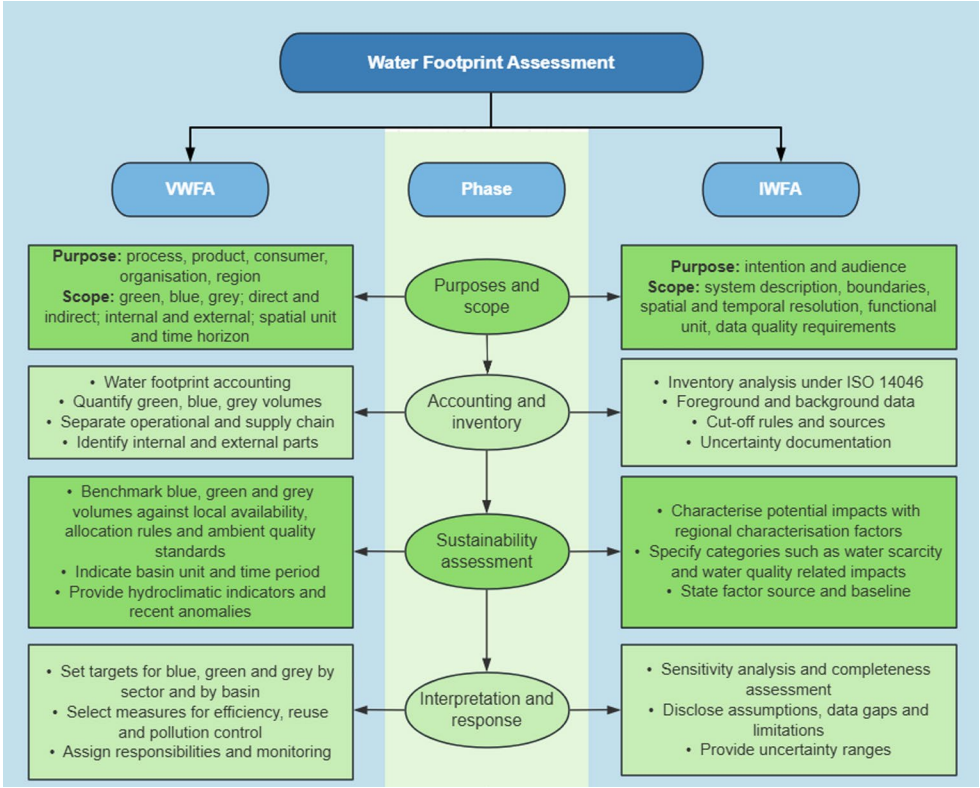


Fig. 4 Comparison of the VWFA and IWFA approaches for WF analysis (adapted from The Water Footprint Assessment Manual and ISO 14046 [7, 13])

consumption and water environment pollution on human health and ecosystems through the cause-effect chain. The two methods focus on different goals, but both can be used to study WF [45].

The current primary methodology for calculating WFs follows the WFN framework. Conceptually, the total WF of a product or region is the sum of the green, blue and grey components within the defined area [7]. At the sectoral level, a regional WF can also be expressed as the sum of the agricultural, industrial and domestic components [7, 98].

From a global perspective, the VWFA method of the WFN integrates a range of disciplines, including hydrology, environmental science and economics, to provide a scientific basis for water resources management [37]. It operationalises this integration by coupling quality standards with supply chain and trade data, and by linking process coefficients to input-output tables to attribute green, blue and grey volumes along production and consumption pathways. This approach emphasizes cross-border water management and international linkages, including transboundary flows within shared basins and the international movement of virtual water through trade.

The calculation of regional WFs is commonly described using two complementary approaches [40]. The top-down approach relies on supply-use or input-output tables, trade statistics and sectoral water coefficients that follow the accounting practices of statistical agencies. Government perspective here means the data structure and aggregation follow official economic accounts and trade records. It assigns virtual water imports and exports explicitly and attributes the remaining internal footprint to sectors and final demand categories [99]. The bottom-up approach uses household surveys, metered use and product specific coefficients to add up residents' direct use and the indirect use embodied in purchased goods and services.

At the sectoral level, the top-down results are useful for planning and budgeting units because they link footprints to economic accounts and trade flows; the bottom-up results are useful for city and community programmes because they link footprints to household behaviour and service provision [40]. Both approaches can be made comparable if they share the same spatial and temporal boundaries, use consistent definitions of green, blue and grey water and draw coefficients from a harmonised source [89, 100].

Most tools currently employed for WF accounting in agriculture are model-based, including the CropWat8.0 crop irrigation model and the AquaCrop growth model, which were developed by the Food and Agriculture Organisation (FAO) of the United Nations [101]. Both models are capable of calculating the green and blue WFs associated with crop production [37]. Furthermore, the sustainability of blue and green water in regional agricultural systems has been evaluated through the SWAT hydrological model, developed

by the United States Department of Agriculture [35]. The GEPIC model, which is based on the Environmental Policy Integrated Climate Model (EPIC), simulates crop growth by inputting relevant climate and soil data to calculate the green WF in crop production [102, 103]. In addition to the calculation of crops, the regional WF of a country or province can be calculated based on the Multi-Regional Input-Output Model (MRIO) to determine the blue and grey WFs [104].

Data availability and consistency remain a primary constraint for WF calculations [105]. Core inputs are typically drawn from hydrometeorological observations, agricultural and industrial statistics, basin and utility records, remote sensing products and multi-regional input-output tables [40]. These sources differ in latency. Hydrometeorological and utility data are updated frequently, whereas national accounts, land use statistics and input-output tables often lag by one to three years. Quality control should include documented metadata, reconciliation of withdrawals, returns and storage at the basin scale, cross checks among independent sources, and transparent treatment of missing values [100, 106]. Model versions differ across the literature, so the model and version used should be reported. When updates modify algorithms or coefficients in ways that can change evapotranspiration, yields or flows, a reanalysis is recommended for decision relevant applications. Use of any model should be justified by demonstrating skill against observations, transparent parameterization and forcing data, sensitivity and uncertainty analysis, and spatial and temporal resolution that matches the assessment boundary.

Implications of Water Footprint

In recent years, research into WFs has increasingly concentrated on the impact of water usage on the environment. This shift reflects a growing awareness that water is not only an economic resource, but also an important component of ecosystems and human well-being. Some studies examine links between water footprints and environmental degradation, including depletion of basin water stocks, biodiversity loss, and water pollution [107]. Here, depletion refers to a persistent reduction in available surface water and groundwater caused by withdrawals and consumptive use that exceed renewable inflows and recovery. This is distinct from meteorological drought, which reflects a temporary climate driven shortage. Other studies analyse environmental sustainability through the grey WFs [108]. Trends in grey WF differ by basin and sector. Increases can be driven by higher pollutant loads, stricter ambient standards or lower background assimilative capacity, and declines occur where treatment coverage and removal efficiency rise or where cleaner production reduces loads [40]. Recycled water that displaces discharge reduces the grey WF to the extent that treated effluent is reused and no longer requires dilution in the receiving water. To maintain accounting consistency, the blue water used for reuse is recorded under consumptive use, while the avoided grey volume is reported with stated assumptions on treatment performance, reuse fraction and return flow [100].

Hoekstra and Hung [47] further developed the concept of virtual water trade within the framework of WF analysis. Their study revealed the extent to which water-scarce countries rely on imports of water-intensive products, effectively outsourcing their water needs to other regions. This research highlights the geopolitical implications of water use and water trade, especially in the context of global water scarcity and food security [109]. Studies of virtual water resource trade have also revealed the environmental and social impacts of such trade. For example, exporting countries often bear the environmental costs of water-intensive agriculture, including groundwater depletion and ecosystem degradation, while importing countries benefit from these resources without experiencing the same degree of environmental stress [110]. For China, most of the water used for product production and service consumption is taken domestically, while exporting virtual water to other countries, which will increase China's water resource pressure [111]. Precipitation and runoff vary strongly across regions and across seasons. Equity should therefore be interpreted as fair outcomes that are conditional in hydroclimatic context, rather than identical volumes everywhere. In dry basins the feasible mix of measures differs from that in wet basins and may rely more on storage, demand management, reuse and trade, whereas in wet basins the emphasis may be on quality control and flood risk [99]. These differences exist within countries as well. In China, contrasts between humid river basins and arid inland basins make uniform targets costly to achieve [78]. These problems have prompted calls for more equitable and sustainable water resource management measures at the global level.

In addition, WF analysis has been employed to evaluate the social consequences of water use, particularly in regions where water scarcity represents a significant threat to life and human well-being. For example, a study has explored the impacts of water-intensive agriculture on local communities, including conflicts over water rights and the displacement of smallholder farmers [112]. Water availability and sanitation are necessary conditions for social equity, but they are not sufficient. The contribution of WF analysis to sustainable development is clearest when linked to specific SDG targets. It can inform SDG 6 (Clean Water and Sanitation) by relating pollutant loads to receiving water standards and by reporting consumptive use and efficiency by sector, SDG 2 (Zero Hunger) by assessing production

footprints against agroecological conditions in food systems, and SDG 12 (Responsible Consumption and Production) by linking consumption footprints to sourcing patterns and waste reduction. Other studies have quantified WF of food consumption patterns, thereby providing a basis for the development of sustainable dietary policies [113]. Comparing WFs of consumption with WF of production can reveal mismatches between local food supply and demand. This provides a basis to harmonise sourcing, storage and substitution while assessing distributional effects under the local hydroclimatic context. Equity in this review refers to access, outcomes and participation. WF informs these discussions, yet it is not a distributive mechanism and cannot equalise access across different hydroclimatic settings. Interpretation is paired with hydroclimatic indicators such as long-term precipitation, aridity and seasonal variability, and we avoid cross region rankings that ignore climate context.

Despite the advancements made in WF research over recent years, numerous challenges and research gaps remain. Firstly, the extant WF models remain inadequate in addressing the complexities of ecosystems and the impact of climate change. Dalin et al. [114] highlighted the necessity of integrating climate change variables into WF calculations and advocated for a more interdisciplinary approach to research. Secondly, the utilisation WF in the context of policymaking remains inadequate. Rosa et al. [115] highlighted that, despite the widespread acceptance of WF as a means of measuring water resource utilisation efficiency, its implementation in actual policies and management strategies remains a significant challenge, particularly in countries with considerable potential for sustainable expansion, such as India and China. A further limitation is the sensitivity of footprint metrics to hydroclimatic setting and to investment constraints, which should be reported with results [116]. Policy relevance also varies by production system. In very arid rangelands with high rainfall variability, where livelihoods depend on mobile pastoralism and opportunistic use of green water, product-based footprints are not always a useful guide for allocation and management [117]. In such settings, effective instruments focus on mobility corridors, drought contingency planning, animal health, market access and social protection rather than on reducing a numerical footprint. Our screened corpus contains few evaluations of water footprint use in these contexts, which we identify as a research gap.

From the perspective of global food security, a high national WF in production can coexist with a large net export of virtual water. Australia provides an illustrative case, where substantial exports support food self-reliance in many importing countries [99]. Several importers are water rich yet land scarce, for example Malaysia and Bangladesh, which constrains their ability to expand domestic production [118]. For exporting countries, cutting crop output does not automatically deliver a net social benefit, and the water saving objectives must be balanced against trade driven food security benefits [119]. To this regard, it is recommended that national water strategies evaluate three indicators in parallel. The net virtual water should be balanced between crops and basin, and the share of exports should be destined for land scarce partners. In addition, the sensitivity of importing markets to supply shortfalls should be explored.

Policymakers can: (i) shift sourcing towards basins with lower water intensity and reduced scarcity; (ii) employ scarcity-weighted indicators to prioritise substitutions that offer the greatest reduction in water stress; (iii) adjust cropping patterns and industrial water reuse targets in export regions; and (iv) mandate supply-chain disclosure aligned with life-cycle reporting [42, 120]. Efficiency can be measured using various metrics, such as virtual-water intensity (m^3 per tonne), water footprint (WF) per unit value added (m^3 per USD), city- or sector-level WF efficiency indices, and scarcity-weighted WF (m^3-equivalents) to account for basin-specific contexts. These metrics enable the comparison of trade or procurement practices based on a "water per benefit" and "stress per unit" approach. The feasibility of these actions depends on investment capacity and infrastructure readiness, including storage, conveyance, treatment, and reuse systems. Consequently, timelines should be phased and cost aware.

Furthermore, in light of the mounting global pressures on water resources, future research in this field shall place a greater emphasis on the realization of SDG 6, SDG 2, and SDG 12. A recent publication [58] suggested that integrating WF analysis with these goals can support the sustainable management of global water resources.

From a global food security perspective, national production WFs need to be interpreted together with cross-border virtual water flows and trade structures [39]. A study in 2005 showed that conclusions drawn from water footprint numbers alone are incomplete unless the direction of virtual water flows and comparative advantage in trade are analysed in tandem [121]. A reassessment in 2020 proposed separating food security challenges from domestic water supply challenges and introduced composite indices that combine water and land availability across many countries, reinforcing the need to account for land scarcity in importing regions [122]. Global datasets and reviews document sizable net exports of agricultural virtual water that sustain external food supply, which implies that reducing irrigated output in exporting basins can impose risks on import-dependent markets unless alternative land or yield gains exist in destination regions [123]. Evidence from a 2019 global analysis quantified that a significant share of unsustainable irrigation is

virtually exported, linking production shortfalls in exporting regions to risks borne by consumers elsewhere [99, 124]. Several importers face land constraints rather than absolute water scarcity [124]. Assessments for Malaysia describe limited scope to expand cropland under sustainability commitments together with persistent reliance on food imports [125].

Development of Standards and Policies

Given enormous water-related challenges, the uninterrupted advancement and implementation of WF analysis are vital to facilitate the attainment of sustainable utilisation and management of water resources. Currently, the WFN and the International Organisation for Standardisation (ISO) are the primary entities proposing pertinent evaluation standards for the concept of the water footprint. ISO 14046 is developed through consensus by national standards bodies within ISO and is an international voluntary standard. Acceptance occurs when countries adopt it nationally or when organisations reference it in procurement, certification, or reporting. Enforcement is indirect, through policies and programs that cite ISO 14046. The Water Footprint Network Manual serves as widely used guidance rather than a binding standard. The United Nations Environment Programme (UNEP) includes the water footprint in its Green Economy Progress framework, which encourages governments and companies to reduce consumptive use and pollution where they impose environmental or economic costs, and to increase efficiency and reuse [126].

In regions with abundant precipitation and periodic floods, the purpose is to lower treatment and pumping costs, limit pollution loads, reduce energy use and emissions, and strengthen resilience to seasonal variation rather than to create water for export [127]. International exchanges of water occur mainly as virtual water embodied in goods, and this framework does not assume cross border sales of bulk water. In the EU's Product Environmental Footprint (PEF), which is based on LCA, the assessment of a product's WF is included as an important indicator for measuring the environmental impact of the product throughout its life cycle. The objective of the PEF guidelines is to assist companies in reducing the environmental impact of their products and enhancing their market competitiveness through the implementation of a unified assessment methodology [128]. The GRI 303 standard, developed by the Global Reporting Initiative (GRI), guides how companies manage and report on their water use. GRI is an independent standard-setting organisation rather than a United Nations project. Its standards are referenced by UN initiatives such as the UN Global Compact and UNEP guidance, and some countries draw on GRI-based corporate disclosures in national sustainability. This standard addresses part of WF by encouraging transparent reporting in corporate sustainability reports [129].

In addition to the impetus provided by international organisations, national and local governments have begun to focus on the role of WF for society and the environment. The government of the Netherlands employs the water footprint as an evaluation tool and has proposed the implementation of water footprint management within a tiered decision-making framework [130]. China published the national standard GB/T 33859 − 2017, entitled "Environmental Management Water Footprint Principles, Requirements and Guidelines", which was based on the ISO 14046 standard. Other countries have implemented policies or tools that are based on the concept of WF. Examples of such initiatives include the United States Water Sense project and the US Water Footprint Calculator [131], and the Australian Water Efficiency Labelling and Standards (WELS) scheme [132]. This indicates that governments are gradually emphasising the role of WF for water resources management and sustainable development. Policy appraisal should report domestic water outcomes together with external food security consequences and set water saving targets with safeguards that protect staple supplies to land scarce trade partners. The WF governance should be aligned with broader environmental and social governance, including basin planning, food system policies and social safeguards.

Conclusions

The water footprint (WF) has become a widely used tool to support the allocation and management of water resources, attracting substantial interest from scholars in recent years. This review covers publications from 2000 to 2023 and synthesises development trends, geographical distribution, primary research focuses, assessment methodologies, implications, and associated policies and standards.

Publications increased markedly over the review period. Research that began mostly in agriculture has expanded to multiple sectors. Geographically, China has the largest publication count with 588 articles, followed by the United States with 221, in both countries, studies still emphasise agricultural over industrial applications. Interpreting these results also requires a global food security lens. Several countries maintain staple supplies through imports from surplus producing regions with large production footprints and net exports of virtual water. Domestic targets and cropping shifts should therefore be evaluated together with trade dependence and land availability in importing regions.

WF assessment relies mainly on two approaches, the volume-oriented (VWFA) and the impact-oriented (IWFA) method, which offer different but complementary

perspectives on water use and environmental consequences. The two methodological lineages are complementary. The VWFA method provides physically interpretable volumes for allocation and accounting. The IWFA method provides characterised impacts for risk and cross regional trade offs. Because their units and model boundaries differ, dual reporting is recommended with explicit metadata instead of aggregation, together with a small set of context indicators that include hydroclimatic setting, infrastructure readiness, the net virtual water balance and the destination profile of exports.

Data accuracy and reproducibility constrain comparability. Policy uptake varies across settings. The capacity to improve footprints depends on capital and infrastructure. Where storage, conveyance, treatment and reuse can be built, options for efficiency and risk reduction are wider. Where development capital is limited, choices are fewer and should be sequenced with attention to costs and feasibility. Empirical evaluations that link WF results to investment constraints and infrastructure readiness are still limited in the screened literature, which were identified as a knowledge gap.

Future work should strengthen the data foundation. Alignment with SDGs can help contribute to broader global sustainability efforts, ensuring that water management strategies support long-term environmental and societal goals. Further WF studies can highlight the need for fair and sustainable management as climate change and conflicts intensify scarcity.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s41101-025-00467-9.

Acknowledgements We would like to thank the Science and Technology Development Fund of Macao for the financial support on this research.

Author Contributions Visualization, literature review, editing, and major revision were performed by **Zhenna Yi**. Data collection, visualization, and first draft were performed by **Chengyan Li**. Editing, revision, and supervision were performed by **Yahong Dong**. All authors read and approved the final manuscript.

Funding This study was funded by the Science and Technology Development Fund of Macao FDCT (No. 0007/2023/RIB1; No. 0146/2022/A).

Data Availability No datasets were generated or analysed during the current study.

Declarations

Clinical Trial Number Not applicable.

Competing interests The authors declare no competing interests.

References

1. Gleick PH (1993) Water in crisis: A guide to the world's fresh water resources. Oxford University Press, Oxford
2. IPCC (2014) Climate change 2014: Impacts, Adaptation, and Vulnerability. Contribution of working group II to the fifth assessment report of the intergovernmental panel on climate change. IPCC, Cambridge, UK and New York, NY, USA
3. Shiklomanov IA (1993) World Fresh Water Resources. In: Gleick PH (ed) Water in Crisis: A Guide to the World's Fresh Water Resources. Oxford University Press, Oxford, pp 13–24
4. UNESCO World Water Assessment Programme (2020) The United Nations world water development report 2020: water and climate change (WWDR 2020)
5. Hoekstra AY, Hung PQ (2003) Virtual Water Trade: Proceedings of the International Expert Meeting on Virtual Water Trade. 1–244
6. Allan JA (1998) Virtual water: a strategic resource global solutions to regional deficits. Groundwater 36:545–548. https://doi.org/10.1111/j.1745-6584.1998.tb02825.x
7. Hoekstra A, Chapagain AK, Aldaya MM, Mekonnen MM (2012) The water footprint assessment manual: Setting the global standard. Routledge
8. Mekonnen MM, Hoekstra AY (2011) The water footprint of humanity. Proc Natl Acad Sci 108:3232–3237. https://doi.org/10.1073/pnas.1109936108
9. Morikawa M, Morrison J, Gleick PH (2019) Water scarcity challenges to business. Nat Clim Change 9:335–337. https://doi.org/10.1038/s41558-019-0466-2
10. Pfister S, Bayer P (2014) Monthly water stress: an indicator for comparing freshwater impacts of global products. Hydrol Earth Syst Sci 18:2325–2335. https://doi.org/10.5194/hess-18-2325-2014
11. Allan JA (1993) Fortunately there are substitutes for water otherwise our hydro-political futures would be impossible. Priorities Water Resour Alloc Manag 13:26
12. Chapagain AK, Hoekstra AY (2008) The global component of freshwater demand and supply: an assessment of virtual water flows between nations as a result of trade in agricultural and industrial products. Water Int 33:19–32. https://doi.org/10.1080/02508060801927812
13. ISO (2014) ISO 14046: 2014 Environmental management – water footprint – principles and guidelines. ISO, Geneva, Switzerland
14. Zhang Y, Huang K, Yu Y, Yang B (2017) Mapping of water footprint research: a bibliometric analysis. J Clean Prod 142:314–327. https://doi.org/10.1016/j.jclepro.2016.10.177
15. Chapagain AK, Hoekstra AY (2004) Water Footprints of Nations. UNESCO-IHE, Delft, The Netherlands
16. Mekonnen MM, Hoekstra AY (2010) The Green, Blue and Grey Water Footprint of Crops and Derived Crop Products. UNESCO-IHE, Delft, The Netherlands
17. Zhuo L, Hoekstra AY, Wang L (2016) The water footprint and virtual water trade of China. Water Resour Manage 30:3951–3965. https://doi.org/10.1007/s11269-016-1407-3
18. Hogeboom R (2020) The water footprint concept and water's grand environmental challenges. One Earth 2(3):218–222
19. Zhu Y, Jiang S, Han X et al (2019) A bibliometrics review of water footprint research in China: 2003–2018. Sustainability 11:5082. https://doi.org/10.3390/su11185082
20. Liu Y, Yang H, Wang H, Yang D (2017) Water footprint assessment at the global scale. J Clean Prod 165:1278–1287. https://doi.org/10.1016/j.jclepro.2017.07.196
21. Pfister S, Koehler A, Hellweg S (2009) Assessing the environmental impacts of freshwater consumption in LCA. Environ Sci Technol 43:4098–4104. https://doi.org/10.1021/es802423e

22. Gerbens-Leenes W, Hoekstra AY, van der Meer TH (2009) The water footprint of bioenergy. Proc Natl Acad Sci U S A 106:10219–10223. https://doi.org/10.1073/pnas.0812619106
23. Ercin AE, Hoekstra AY (2014) Water footprint scenarios for 2050: a global analysis. Environ Int 64:71–82. https://doi.org/10.1016/j.envint.2013.11.019
24. Mekonnen MM, Hoekstra AY (2012) A global assessment of the water footprint of farm animal products. Ecosystems 15:401–415. https://doi.org/10.1007/s10021-011-9517-8
25. Hoekstra AY (2019) The water footprint of modern consumer society, 2nd edn. Routledge, London
26. Vanham D, Bidoglio G (2013) The water footprint of Milan. Water Sci Technol 69:789–795. https://doi.org/10.2166/wst.2013.759
27. Chapagain AK, Hoekstra AY (2007) The water footprint of coffee and tea consumption in the Netherlands. Ecol Econ 64:109–118
28. Hoekstra AY (2017) Water footprint assessment: evolvement of a new research field. Water Resour Manag 31:3061–3081. https://doi.org/10.1007/s11269-017-1618-5
29. Mekonnen MM, Hoekstra AY (2016) Four billion people facing severe water scarcity. Sci Adv 2:e1500323. https://doi.org/10.1126/sciadv.1500323
30. Chapagain AK, Hoekstra AY (2011) The blue, green and grey water footprint of rice from production and consumption perspectives. Ecol Econ 70:749–758. https://doi.org/10.1016/j.ecolecon.2010.11.012
31. Hoekstra AY, Wiedmann TO (2014) Humanity's unsustainable environmental footprint. Science 344:1114–1117. https://doi.org/10.1126/science.1248365
32. Wichelns D (2010) Virtual water: a helpful perspective, but not a sufficient policy criterion. Water Resour Manage 24:2203–2219. https://doi.org/10.1007/s11269-009-9547-6
33. Vanham D, Bidoglio G (2013) A review on the indicator water footprint for the EU28. Ecol Indic 26:61–75. https://doi.org/10.1016/j.ecolind.2012.10.021
34. Li M, Fu Q, Singh VP et al (2020) Sustainable management of land, water, and fertilizer for rice production considering footprint family assessment in a random environment. J Clean Prod 258:120785. https://doi.org/10.1016/j.jclepro.2020.120785
35. Bigdeli Nalbandan R, Delavar M, Abbasi H, Zaghiyan MR (2023) Model-based water footprint accounting framework to evaluate new water management policies. J Clean Prod 382:135220. https://doi.org/10.1016/j.jclepro.2022.135220
36. Zhao D, Liu J, Sun L et al (2021) Quantifying economic-social-environmental trade-offs and synergies of water-supply constraints: an application to the capital region of China. Water Res 195:116986
37. Mekonnen MM, Hoekstra AY (2011) The green, blue and grey water footprint of crops and derived crop products. Hydrol Earth Syst Sci 15:1577–1600. https://doi.org/10.5194/hess-15-1577-2011
38. Ercin AE, Aldaya MM, Hoekstra AY (2011) Corporate water footprint accounting and impact assessment: the case of the water footprint of a sugar-containing carbonated beverage. Water Resour Manag 25:721–741
39. Hoekstra AY, Mekonnen MM (2012) The water footprint of humanity. Proc Natl Acad Sci 109:3232–3237
40. Aldaya MM, Chapagain AK, Hoekstra AY, Mekonnen MM (2012) The water footprint assessment manual: setting the global standard. Routledge
41. Hoekstra AY, Chapagain AK (2007) Water footprints of nations: water use by people as a function of their consumption pattern. Water Resour Manag 21:35–48
42. Boulay A-M, Bare J, Benini L et al (2018) The WULCA consensus characterization model for water scarcity footprints: assessing impacts of water consumption based on available water remaining (AWARE). Int J Life Cycle Assess 23:368–378
43. Pierrat É, Laurent A, Dorber M et al (2023) Advancing water footprint assessments: combining the impacts of water pollution and scarcity. Sci Total Environ 870:161910
44. Pfister S, Boulay A-M, Berger M et al (2017) Understanding the LCA and ISO water footprint: a response to Hoekstra (2016) "A critique on the water-scarcity weighted water footprint in LCA." Ecol Indic 72:352–359. https://doi.org/10.1016/j.ecolind.2016.07.051
45. Hoekstra AY (2016) A critique on the water-scarcity weighted water footprint in LCA. Ecol Indic 66:564–573. https://doi.org/10.1016/j.ecolind.2016.02.026
46. WWAP (United Nations World Water Assessment Programme) (2016) The United Nations World Water Development Report 2016: Water and Jobs. Unesco, Paris
47. Hoekstra AY, Hung PQ (2005) Globalisation of water resources: international virtual water flows in relation to crop trade. Glob Environ Change 15:45–56. https://doi.org/10.1016/j.gloenvcha.2004.06.004
48. Liu J, Wiberg D, Zehnder AJB, Yang H (2007) Modeling the role of irrigation in winter wheat yield, crop water productivity, and production in China. Irrig Sci 26:21–33. https://doi.org/10.1007/s00271-007-0069-9
49. Lovarelli D, Bacenetti J, Fiala M (2016) Water footprint of crop productions: a review. Sci Total Environ 548:236–251. https://doi.org/10.1016/j.scitotenv.2016.01.022
50. Zhao H, Qu S, Guo S et al (2019) Virtual water scarcity risk to global trade under climate change. J Clean Prod 230:1013–1026. https://doi.org/10.1016/j.jclepro.2019.05.114
51. Zhao Y, Ding D, Si B et al (2019) Temporal variability of water footprint for cereal production and its controls in Saskatchewan, Canada. Sci Total Environ 660:1306–1316. https://doi.org/10.1016/j.scitotenv.2018.12.410
52. Cao X, Huang X, Huang H et al (2018) Changes and driving mechanism of water footprint scarcity in crop production: a study of Jiangsu Province, China. Ecol Indic 95:444–454. https://doi.org/10.1016/j.ecolind.2018.07.059
53. Li J, Liu Z, He C et al (2017) Water shortages raised a legitimate concern over the sustainable development of the drylands of northern China: evidence from the water stress index. Sci Total Environ 590:739–750. https://doi.org/10.1016/j.scitotenv.2017.03.037
54. Munia H, Guillaume JHA, Mirumachi N et al (2016) Water stress in global transboundary river basins: significance of upstream water use on downstream stress. Environ Res Lett 11:014002. https://doi.org/10.1088/1748-9326/11/1/014002
55. Nazari R, Ramezani Etedali H, Nazari B, Collins B (2020) The impact of climate variability on water footprint components of rainfed wheat and barley in the Qazvin province of Iran*. Irrig Drain 69:826–843. https://doi.org/10.1002/ird.2487
56. Bocchiola D (2015) Impact of potential climate change on crop yield and water footprint of rice in the Po Valley of Italy. Agric Syst 139:223–237. https://doi.org/10.1016/j.agsy.2015.07.009
57. Huang Y, Han R, Qi J et al (2022) Health risks of industrial wastewater heavy metals based on improved grey water footprint model. J Clean Prod 377:134472. https://doi.org/10.1016/j.jclepro.2022.134472
58. Weerasooriya RR, Liyanage LPK, Rathnappriya RHK et al (2021) Industrial water conservation by water footprint and sustainable development goals: a review. Environ Dev Sustain 23:12661–12709. https://doi.org/10.1007/s10668-020-01184-0
59. Ren X, Bai X, Hu M et al (2019) Water footprint assessment of textile enterprise based on ISO14046. Desalin Water Treat 168:216–223. https://doi.org/10.5004/dwt.2019.24203

60. Zhang Y, Wu XY, Wang LL, Ding XM (2014) The industrial water footprint of zippers. Water Sci Technol 70:1025–1031. https://doi.org/10.2166/wst.2014.323
61. Wang L, Ding X, Wu X, Yu J (2013) Textiles industrial water footprint: methodology and study. JSIR 7211
62. Zhu Y, Wang J, He G et al (2022) Water use characteristics and water footprints of China's oil and gas production. Resour Conserv Recycl 184:106428. https://doi.org/10.1016/j.resconrec.2022.106428
63. Chini CM, Logan LH, Stillwell AS (2020) Grey water footprints of U.S. thermoelectric power plants from 2010–2016. Adv Water Resour 145:103733. https://doi.org/10.1016/j.advwatres.2020.103733
64. Fialkiewicz W, Burszta-Adamiak E, Kolonko-Wiercik A et al (2018) Simplified direct water footprint model to support urban water management. Water 10:630. https://doi.org/10.3390/w10050630
65. Konar M, Marston L (2020) The water footprint of the United States. Water 12:3286. https://doi.org/10.3390/w12113286
66. Chini CM, Konar M, Stillwell AS (2017) Direct and indirect urban water footprints of the United States. Water Resour Res 53:316–327. https://doi.org/10.1002/2016WR019473
67. Wiedmann T, Allen C (2021) City footprints and SDGs provide untapped potential for assessing city sustainability. Nat Commun 12:3758
68. Shi S, Yin J (2021) Global research on carbon footprint: a scientometric review. Environ Impact Assess Rev 89:106571. https://doi.org/10.1016/j.eiar.2021.106571
69. Chen C (2006) CiteSpace II: detecting and visualizing emerging trends and transient patterns in scientific literature. J Am Soc Inf Sci Technol 57:359–377. https://doi.org/10.1002/asi.20317
70. Hoekstra AY (2017) Water footprint assessment: evolvement of a new research field. Water Resour Manag 31:3061–3081
71. Zhang Y, Huang K, Yu Y, Yang B (2017) Mapping of water footprint research: a bibliometric analysis during 2006–2015. J Clean Prod 149:70–79
72. Shneider AM (2009) Four stages of a scientific discipline; four types of scientist. Trends Biochem Sci 34:217–223. https://doi.org/10.1016/j.tibs.2009.02.002
73. Gómez-Llanos E, Durán-Barroso P, Matías-Sánchez A (2018) Management effectiveness assessment in wastewater treatment plants through a new water footprint indicator. J Clean Prod 198:463–471. https://doi.org/10.1016/j.jclepro.2018.07.062
74. Yu F, Wang Y, Liu X et al (2023) Driving forces and variation in water footprint before and after the COVID-19 lockdown in Fujian Province of China. J Clean Prod 402:136696. https://doi.org/10.1016/j.jclepro.2023.136696
75. Li C, Xu M, Wang X, Tan Q (2018) Spatial analysis of dual-scale water stresses based on water footprint accounting in the Haihe River Basin, China. Ecol Indic 92:254–267. https://doi.org/10.1016/j.ecolind.2017.02.046
76. Feng Q, Sun Y, Li X et al (2024) A bibliometric analysis of water footprint in China and abroad: research hotspots and frontiers. Water Sci Technol 90:1759–1783
77. Liu J, Zhao D, Mao G et al (2020) Environmental sustainability of water footprint in mainland China. Geogr Sustain 1:8–17. https://doi.org/10.1016/j.geosus.2020.02.002
78. Ma T, Sun S, Fu G et al (2020) Pollution exacerbates China's water scarcity and its regional inequality. Nat Commun 11:650
79. Wang W, Wu F, Yu H, Wang X (2023) Assessing the effectiveness of intervention policies for reclaimed water reuse in China considering multi-scenario simulations. J Environ Manage 335:117519
80. Zhang C-Y, Oki T (2023) Water pricing reform for sustainable water resources management in China's agricultural sector. Agric Water Manage 275:108045
81. Zhuo L, Mekonnen MM, Hoekstra AY (2016) Benchmark levels for the consumptive water footprint of crop production for different environmental conditions: a case study for winter wheat in China. Hydrol Earth Syst Sci 20:4547–4559
82. Hogeboom RJ, De Bruin D, Schyns JF et al (2020) Capping human water footprints in the world's river basins. Earths Future 8:e2019EF001363. https://doi.org/10.1029/2019EF001363
83. Abbas F, Al-Naemi S, Farooque AA, Phillips M (2023) A review on the water dimensions, security, and governance for two distinct regions. Water 15:208
84. Zhang Z, Shan Y, Zhao D et al (2024) City level water withdrawal and scarcity accounts of China. Sci Data 11:449
85. Huang C, Chen T, Yi H et al (2017) Collaborative environmental governance, inter-agency cooperation and local water sustainability in China. Sustainability 9:2305
86. Ouyang J, Zhang K, Wen B, Lu Y (2020) Top-down and bottom-up approaches to environmental governance in china: evidence from the river chief system (RCS). Int J Environ Res Public Health 17:7058
87. Chapagain AK, Hoekstra AY, Savenije HH, Gautam R (2006) The water footprint of cotton consumption: an assessment of the impact of worldwide consumption of cotton products on the water resources in the cotton producing countries. Ecol Econ 60:186–203
88. Novoa V, Ahumada-Rudolph R, Rojas O et al (2019) Understanding agricultural water footprint variability to improve water management in Chile. Sci Total Environ 670:188–199. https://doi.org/10.1016/j.scitotenv.2019.03.127
89. Velasco-Muñoz JF, Aznar-Sánchez JA, Belmonte-Ureña LJ, Román-Sánchez IM (2018) Sustainable water use in agriculture: a review of worldwide research. Sustainability 10:1084. https://doi.org/10.3390/su10041084
90. Sun Y, Wang Z, Lee L-C et al (2022) A bibliometrics review of hotspots in water footprint research based on co-words network analysis. Front Environ Sci 10:1027936. https://doi.org/10.3389/fenvs.2022.1027936
91. Ding Y, Li Y, Zheng H et al (2022) Mapping water, energy and carbon footprints along urban agglomeration supply chains. Earth 's Future 10:e2021EF002225. https://doi.org/10.1029/2021EF002225
92. Zeng Z, Liu J, Koeneman PH et al (2012) Assessing water footprint at river basin level: a case study for the Heihe river basin in Northwest China. Hydrol Earth Syst Sci 16:2771–2781. https://doi.org/10.5194/hess-16-2771-2012
93. Ruiz-Pérez MR, Alba-Rodríguez MD, Marrero M (2022) Evaluation of water footprint of urban renewal projects. Case study in Seville, Andalusia. Water Res 221:118715. https://doi.org/10.1016/j.watres.2022.118715
94. Paterson W, Rushforth R, Ruddell BL et al (2015) Water footprint of cities: a review and suggestions for future research. Sustainability 7:8461–8490. https://doi.org/10.3390/su7078461
95. Mahjabin T, Garcia S, Grady C, Mejia A (2018) Large cities get more for less: water footprint efficiency across the US. PLoS One 13:e0202301. https://doi.org/10.1371/journal.pone.0202301
96. Zhao R, He H, Zhang N (2015) Regional water footprint assessment: a case study of Leshan City. Sustainability 7:16532–16547. https://doi.org/10.3390/su71215829
97. Baccour S, Goelema G, Kahil T et al (2024) Water quality management could halve future water scarcity cost-effectively in the Pearl river basin. Nat Commun 15:5669
98. Cai J, He Y, Xie R, Liu Y (2020) A footprint-based water security assessment: an analysis of Hunan province in China. J Clean Prod 245:118485. https://doi.org/10.1016/j.jclepro.2019.118485
99. d'Odorico P, Carr J, Dalin C et al (2019) Global virtual water trade and the hydrological cycle: patterns, drivers, and socio-environmental impacts. Environ Res Lett 14:53001

100. Karimi P, Bastiaanssen WG, Molden D (2013) Water accounting plus (WA+)–a water accounting procedure for complex river basins based on satellite measurements. Hydrol Earth Syst Sci 17:2459–2472
101. Steduto P, Hsiao TC, Raes D, Fereres E (2009) AquaCrop—the FAO crop model to simulate yield response to water: I. concepts and underlying principles. Agron J 101:426–437
102. Yin Y, Gao Y, Lin D et al (2021) Mapping the Global-Scale maize drought risk under climate change based on the GEPIC-Vulnerability-Risk model. Int J Disaster Risk Sci 12:428–442. https://doi.org/10.1007/s13753-021-00349-3
103. Liu J, Williams JR, Zehnder AJB, Yang H (2007) GEPIC – modelling wheat yield and crop water productivity with high resolution on a global scale. Agric Syst 94:478–493. https://doi.org/10.1016/j.agsy.2006.11.019
104. Sturla G, Ciulla L, Rocchi B (2023) Natural and social scarcity in water Footprint: a multiregional input–output analysis for Italy. Ecol Indic 147:109981. https://doi.org/10.1016/j.ecolind.2023.109981
105. Chenoweth J, Hadjikakou M, Zoumides C (2014) Quantifying the human impact on water resources: a critical review of the water footprint concept. Hydrol Earth Syst Sci 18:2325–2342
106. Esen SE, Hein L (2020) Development of SEEA water accounts with a hydrological model. Sci Total Environ 737:140168
107. Falkenmark M, Rockström J (2006) The new blue and green water paradigm: breaking new ground for water resources planning and management. J Water Resour Plann Manage 132:129–132. https://doi.org/10.1061/(ASCE)0733-9496(2006)132:3(129)
108. Xu C, Liu Y, Fu T (2022) Spatial-temporal evolution and driving factors of grey water footprint efficiency in the Yangtze River Economic Belt. Sci Total Environ 844:156930. https://doi.org/10.1016/j.scitotenv.2022.156930
109. Yang H, Zehnder A (2007) Virtual water": an unfolding concept in integrated water resources management. Water Resour Res. https://doi.org/10.1029/2007WR006048
110. Verma S, Kampman DA, van der Zaag P, Hoekstra AY (2009) Going against the flow: a critical analysis of inter-state virtual water trade in the context of India's National River Linking Program. Physics and Chemistry of the Earth, Parts A/B/C 34:261–269. https://doi.org/10.1016/j.pce.2008.05.002
111. Wang H, Yang Y (2018) Trends and consumption structures of China's blue and grey water footprint. Water 10:494. https://doi.org/10.3390/w10040494
112. Aldaya MM, Llamas MR (2008) Water footprint analysis for the guadiana river basin
113. García-Herrero L, Gibin D, Damiani M et al (2023) What is the water footprint of EU food consumption? A comparison of water footprint assessment methods. J Clean Prod 415:137807. https://doi.org/10.1016/j.jclepro.2023.137807
114. Dalin C, Suweis S, Konar M et al (2012) Modeling past and future structure of the global virtual water trade network. Geophys Res Lett. https://doi.org/10.1029/2012GL053871
115. Rosa L, Rulli MC, Davis KF et al (2018) Closing the yield gap while ensuring water sustainability. Environ Res Lett 13:104002. https://doi.org/10.1088/1748-9326/aadeef
116. Damkjaer S, Taylor R (2017) The measurement of water scarcity: defining a meaningful indicator. Ambio 46:513–531
117. Doreau M, Corson MS, Wiedemann SG (2012) Water use by livestock: a global perspective for a regional issue? Anim Front 2:9–16
118. Rai R, Zhang Y, Paudel B et al (2017) A synthesis of studies on land use and land cover dynamics during 1930–2015 in Bangladesh. Sustainability 9:1866
119. Dixon PB, Rimmer MT, Wittwer G (2011) Saving the Southern murray-darling basin: the economic effects of a buyback of irrigation water. Econ Rec 87:153–168
120. Hoekstra AY (2017) Water footprint assessment in supply chains. In: Bouchery Y, Corbett CJ, Fransoo JC, Tan T (eds) Sustainable supply chains: a research-based textbook on operations and strategy. Springer International Publishing, Cham, pp 65–85
121. Kumar MD, Bassi N, Singh OP (2021) Rethinking on the methodology for assessing global water and food challenges. In: Tortajada C, Araral E (eds) Global Water Resources: Festschrift in Honour of Asit K. Biswas. Routledge, pp 325–342.
122. Kumar MD, Singh OP (2005) Virtual water in global food and water policy making: is there a need for rethinking? Water Resour Manag 19:759–789
123. Mekonnen MM, Kebede MM, Demeke BW et al (2024) Trends and environmental impacts of virtual water trade. Nat Rev Earth Environ 5:890–905
124. Rosa L, Chiarelli DD, Tu C et al (2019) Global unsustainable virtual water flows in agricultural trade. Environ Res Lett 14:114001
125. DeClerck FA, Jones S, Consortium F, Others (2019) Pathways to sustainable land-use and food systems: 2019 report of the FABLE consortium
126. UNEP (2020) Feasibility study on strengthening the environmental footprints and planetary boundaries concepts within the green economy progress measurement framework
127. Intergovernmental Panel on Climate Change (2007) Impacts, adaptation and vulnerability. Genebra Suíça
128. Manfredi S, Allacker K, Pelletier N et al (2012) Product environmental footprint (PEF) guide
129. Global Reporting Initiative, GRI (2018) GRI 303: Water and Effluents 2018
130. Simons G Strengths and Limitations of Water Footprints: a Literature Review
131. U.S. Environmental Protection Agency (2025) WaterSense. https://www.epa.gov/watersense. Accessed 20 Oct 2025
132. Australian Government Department of Climate Change, Energy, the Environment and Water (2025) Water efficiency labelling and standards (WELS) scheme. https://www.waterrating.gov.au/. Accessed 20 Oct 2025

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.